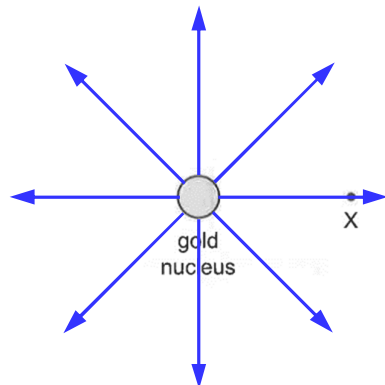


$\approx 5.5 \text{ V}$

23

D



Gold nucleus is positively charged. Electric field lines radiate uniformly outwards from the nucleus.

The direction of the electric field strength at X is to the right.

The electric charge of an α -particle is $2e$.

Magnitude of electric field strength,

$$\begin{aligned} E &= \frac{F}{q} = \frac{F}{2e} \\ &= \frac{35}{2 \times 1.60 \times 10^{-19}} \\ &= 1.09 \times 10^{20} \\ &\approx 1.1 \times 10^{20} \text{ N C}^{-1} \end{aligned}$$

24

D

The positive charge is stationary inside the uniform magnetic field. Since